

ORF309 S2025 Midterm2

April 2025

Problem 1. Alice is planning a trip to California and is observing the price movement of flights that depart from Newark Liberty International Airport (she is quite flexible about travel dates). If the price goes down to 200 dollars, she will purchase the flight ticket and start packing her belongings. However, if the price goes up to 900 dollars, it will be so expensive that she will give up and cancel the travel plan. For each hour, the price either increases by 2 dollars with probability $\frac{1}{3}$ or decreases by 1 dollar with probability $\frac{2}{3}$. We assume the price movements within different hours are mutually independent.

- (1) Assume the current price is 500 dollars. What is the expected price after 24 hours?
- (2) Let $p(i)$ denote the probability that Alice gives up traveling before purchasing the ticket, given the current price is i dollars. Use first-step analysis to rigorously derive a recursive formula for $p(i)$, $200 < i < 900$.
- (3) Let $\mu(i)$ denote the expected number of hours until Alice makes any decision (either she purchases the ticket or gives up traveling). Use first-step analysis to rigorously derive a recursive formula for $\mu(i)$, $200 < i < 900$.

Solution. Let X_n be the price increment of the n -th hour, and S_n be the price after n hours. Then

$$S_n = S_0 + \sum_{i=1}^n X_i$$

is a random walk.

- (1) It is easy to calculate that

$$\mathbb{E}[X_n] = 2\mathbb{P}[X_n = 2] + (-1)\mathbb{P}[X_n = -1] = 2 \times \frac{1}{3} - 1 \times \frac{2}{3} = 0.$$

By the linearity of expectation,

$$\mathbb{E}[S_n] = S_0 + \sum_{i=1}^n \mathbb{E}[X_i] = S_0 + 0 = S_0.$$

- (2) Let

$$T := \min\{n \geq 0 : S_n = 900 \text{ or } S_n = 200\}$$

be the time that Alice makes any decision. Then the event that she gives up traveling can be expressed as $\{S_T = 900\}$. By the law of total probability,

$$p(i) = \mathbb{P}[S_T = 900 \mid S_0 = i]$$

$$\begin{aligned}
&= \mathbb{P}[S_T = 900 \mid S_1 = i + 2, S_0 = i] \cdot \mathbb{P}[S_1 = i + 2 \mid S_0 = i] \\
&\quad + \mathbb{P}[S_T = 900 \mid S_1 = i - 1, S_0 = i] \cdot \mathbb{P}[S_1 = i - 1 \mid S_0 = i] \\
&= \mathbb{P}[S_T = 900 \mid S_0 = i + 2] \cdot \mathbb{P}[X_1 = 2] \\
&\quad + \mathbb{P}[S_T = 900 \mid S_0 = i - 1] \cdot \mathbb{P}[X_1 = -1] \\
&= \frac{1}{3}p(i + 2) + \frac{2}{3}p(i - 1).
\end{aligned}$$

(3) By the law of total expectation,

$$\begin{aligned}
\mu(i) &= \mathbb{E}[T \mid S_0 = i] \\
&= \mathbb{E}[T \mid S_1 = i + 2, S_0 = i] \cdot \mathbb{P}[S_1 = i + 2 \mid S_0 = i] \\
&\quad + \mathbb{E}[T \mid S_1 = i - 1, S_0 = i] \cdot \mathbb{P}[S_1 = i - 1 \mid S_0 = i] \\
&= (1 + \mathbb{E}[T \mid S_0 = i + 2]) \cdot \mathbb{P}[X_1 = 2] \\
&\quad + (1 + \mathbb{E}[T \mid S_0 = i - 1]) \cdot \mathbb{P}[X_1 = -1] \\
&= 1 + \frac{1}{3}\mu(i + 2) + \frac{2}{3}\mu(i - 1).
\end{aligned}$$

□

Problem 2. Flights at Newark Liberty International Airport often experience unpredictable delays and, according to the Federal Aviation Administration, the largest cause of air traffic delays in the National Airspace System is weather. Let the weather condition be quantitatively modeled as a single index (called the weather index) Z , and a higher value of Z means better weather. Suppose the weather index Z is a random variable that is exponentially distributed with mean 1. Given this index Z , the flight delay time T (in hours) will be exponentially distributed with mean $1/Z$. In other words,

$$\mathbb{P}[T > t \mid Z = z] = e^{-zt}, \quad \forall t, z > 0.$$

- (1) Determine the joint probability density function for the pair of random variables (T, Z) .
- (2) Suppose the weather condition is better than average (that is, $Z > 1$). Given this information, what is the probability that the flight will be delayed by more than 2 hours?
- (3) Suppose the current flight is delayed by exactly 2 hours. Given this information, what is the probability that the weather condition is better than average (that is, $Z > 1$)?

Solution. (1) We know that the marginal probability density function of Z is given by

$$f_Z(z) = e^{-z} \mathbf{1}_{(0, +\infty)}(z)$$

and that the conditional density of T given Z is given by

$$f_{T|Z}(t|z) = -\frac{d}{dt}\mathbb{P}[T > t \mid Z = z] = ze^{-zt} \mathbf{1}_{(0, +\infty)}(z) \mathbf{1}_{(0, +\infty)}(t).$$

So the joint density of (T, Z) is given by

$$f_{T,Z}(t, z) = f_Z(z) f_{T|Z}(t|z) = ze^{-z(t+1)} \mathbf{1}_{(0, +\infty)}(z) \mathbf{1}_{(0, +\infty)}(t).$$

(2) By the definition of conditional probabilities,

$$\mathbb{P}[T > 2 \mid Z > 1] = \frac{\mathbb{P}[T > 2, Z > 1]}{\mathbb{P}[Z > 1]}.$$

The denominator follows easily from the distribution of Z :

$$\mathbb{P}[Z > 1] = e^{-1}.$$

For the numerator, we condition on Z and apply the law of total probability:

$$\begin{aligned}\mathbb{P}[T > 2, Z > 1] &= \int_1^\infty \mathbb{P}[T > 2 \mid Z = z] \cdot f_Z(z) dz \\ &= \int_1^\infty e^{-2z} \cdot e^{-z} dz = \frac{1}{3}e^{-3}.\end{aligned}$$

As a result,

$$\mathbb{P}[T > 2 \mid Z > 1] = \frac{\mathbb{P}[T > 2, Z > 1]}{\mathbb{P}[Z > 1]} = \frac{\frac{1}{3}e^{-3}}{e^{-1}} = \frac{1}{3}e^{-2}.$$

(3) We first calculate the marginal probability density of T :

$$f_T(t) = \int_0^\infty f_{T,Z}(t, z) dz = \int_0^\infty ze^{-z(t+1)} dz \cdot \mathbf{1}_{(0,+\infty)}(t) = \frac{1}{(t+1)^2} \cdot \mathbf{1}_{(0,+\infty)}(t).$$

Given this, we can calculate the conditional density of Z given T :

$$f_{Z|T}(z|t) = \frac{f_{T,Z}(t, z)}{f_T(t)} = (t+1)^2 ze^{-z(t+1)} \mathbf{1}_{(0,+\infty)}(z) \mathbf{1}_{(0,+\infty)}(t).$$

As a result, taking $t = 2$,

$$\mathbb{P}[Z > 1 \mid T = 2] = \int_1^\infty f_{Z|T}(z|2) dz = \int_1^\infty 3^2 ze^{-3z} dz = 4e^{-3}.$$

□

Problem 3. Many travelers who have landed at Newark Liberty International Airport will take taxis to get to their destinations. At the taxi stop, passengers arrive according to a Poisson process with a rate 2 passengers per minute (they join the queue and wait until a taxi is ready for them), and taxis arrive according to a Poisson process with a rate 1 vehicle per minute (they pick up one passenger each and wait at the stop if there are no waiting passengers). We assume that the arrivals of travelers and taxis are independent and that currently there are no travelers or taxis waiting at the stop.

- (1) What is the probability that at least 10 passengers arrive before the first taxi comes?
- (2) What is the expected time until the first taxi arrives, given that by the time it arrives there are exactly 10 passengers waiting at the stop?
- (3) Accidents, which lead to temporary traffic breakdowns, occur, independently of the arrivals of travelers and vehicles, according to a Poisson process with a rate of once per 100 minutes. What is the probability that at least one passenger is picked up by a taxi before any accident occurs?

Solution. Let $(N_t^P)_{t \geq 0}$ be the Poisson process corresponding to the arrivals of passengers, $(N_t^T)_{t \geq 0}$ be the Poisson process corresponding to the arrivals of taxis, and $(N_t^A)_{t \geq 0}$ be the Poisson process corresponding to the arrivals of accidents. They are independent Poisson processes with rates

$\lambda_P = 2$, $\lambda_T = 1$ and $\lambda_A = \frac{1}{100}$, respectively.

(1) Let $N_t := N_t^P + N_t^T$ be the aggregate process, which is a Poisson process with rate $\lambda_T + \lambda_P = 3$ by the Principle of Superposition. Let $X_i \in \{P, T\}$ be the type of the i -th arrival of $(N_t)_{t \geq 0}$. Then by the Principle of Thinning, $(X_i)_{i \geq 1}$ is a sequence of i.i.d. random variables with distribution

$$\mathbb{P}[X_i = P] = \frac{\lambda_P}{\lambda_P + \lambda_T} = \frac{2}{3}, \quad \mathbb{P}[X_i = T] = \frac{\lambda_T}{\lambda_P + \lambda_T} = \frac{1}{3}.$$

The event that at least 10 passengers arrive before the first taxi comes can be expressed as

$$\{X_i = P, \quad 1 \leq i \leq 10\} = \bigcap_{i=1}^{10} \{X_i = P\}.$$

By the independence of $(X_i)_{i \geq 1}$, its probability is

$$\mathbb{P} \left[\bigcap_{i=1}^{10} \{X_i = P\} \right] = \prod_{i=1}^{10} \mathbb{P}[X_i = P] = \left(\frac{2}{3} \right)^{10}.$$

(2) Let τ_1^T denote the first arrival time of taxis, and let τ_i denote the i -th arrival time of the aggregated process $(N_t)_{t \geq 0}$. According to the Principle of Superposition and Thinning, the two families of random variables $(\tau_i)_{i \geq 1}$ and $(X_i)_{i \geq 1}$ are independent. Note that the event that by the time the first taxi arrives there are exactly 10 passengers waiting at the stop can be expressed as

$$\{X_1 = X_2 = \dots = X_{10} = P, X_{11} = T\},$$

on which $\tau_1^T = \tau_{11}$. As a result, the desired conditional expectation can be evaluated as

$$\begin{aligned} \mathbb{E}[\tau_1^T \mid X_1 = X_2 = \dots = X_{10} = P, X_{11} = T] &= \mathbb{E}[\tau_{11} \mid X_1 = X_2 = \dots = X_{10} = P, X_{11} = T] \\ &= \mathbb{E}[\tau_{11}] = \frac{11}{\lambda_P + \lambda_T} = \frac{11}{3}, \end{aligned}$$

where in the second last equality we use the fact that τ_{11} can be realized as a sum of 11 exponentially distributed random variables with parameter $\lambda_P + \lambda_T$, and in the last equality we use the fact that τ_{11} and the event $\{X_1 = X_2 = \dots = X_{10} = P, X_{11} = T\}$ are independent. $\square$

(3) [Method 1] The event that at least one passenger is picked up by a taxi before any accidents occurs can be translated into that the first arrivals of both passengers and taxis have to occur before the first arrival of accidents. This can be further decomposed into two cases: (a) The first arrival among passengers, taxis and accidents is a passenger, and after that arrival, the first arrival among taxis and accidents is a taxi; (b) The first arrival among passengers, taxis and accidents is a taxi, and after that arrival, the first arrival among passengers and accidents is a passenger. These two events are mutually exclusive, so the desired probability is the sum of their probabilities.

Let's evaluate the probability of the event in case (a). First, aggregating the three Poisson processes and then apply the Principle of Thinning, we see that

$$\mathbb{P}[\{\text{The first arrival among passengers, taxis and accidents is a passenger}\}] = \frac{\lambda_P}{\lambda_P + \lambda_T + \lambda_A}.$$

As Poisson processes have independent increments, anything happens after that first arrival time is independent of anything happens before that first arrival time. As a result,

$$\begin{aligned} & \mathbb{P}[\{\text{after that arrival, the first arrival among taxis and accidents is a taxi}\} \\ & \quad | \{\text{The first arrival among passengers, taxis and accidents is a passenger}\}] \\ &= \mathbb{P}[\{\text{the first arrival among taxis and accidents is a taxi}\}] = \frac{\lambda_T}{\lambda_T + \lambda_A}. \end{aligned}$$

Therefore, the probability of the event in case (a) is

$$\frac{\lambda_P}{\lambda_P + \lambda_T + \lambda_A} \cdot \frac{\lambda_T}{\lambda_T + \lambda_A}.$$

Similarly, the probability of the event in case (b) is

$$\frac{\lambda_T}{\lambda_P + \lambda_T + \lambda_A} \cdot \frac{\lambda_P}{\lambda_P + \lambda_A}.$$

As a result, the desired probability is

$$\frac{\lambda_P}{\lambda_P + \lambda_T + \lambda_A} \cdot \frac{\lambda_T}{\lambda_T + \lambda_A} + \frac{\lambda_T}{\lambda_P + \lambda_T + \lambda_A} \cdot \frac{\lambda_P}{\lambda_P + \lambda_A} = \frac{2}{3.01} \times \frac{1}{1.01} + \frac{1}{3.01} \times \frac{2}{2.01}.$$

[Method 2] Let τ denote the arrival time of the first accident, which is exponentially distributed with parameter $\frac{1}{100}$. The desired event can be expressed as

$$\{N_\tau^P \geq 1, N_\tau^T \geq 1\}.$$

We use the mutual independence and the law of total probability to evaluate

$$\begin{aligned} \mathbb{P}[N_\tau^P \geq 1, N_\tau^T \geq 1] &= \int_0^\infty \mathbb{P}[N_\tau^P \geq 1, N_\tau^T \geq 1 | \tau = t] \mathbb{P}[\tau \in dt] \\ &= \int_0^\infty \mathbb{P}[N_t^P \geq 1, N_t^T \geq 1] \mathbb{P}[\tau \in dt] \\ &= \int_0^\infty \mathbb{P}[N_t^P \geq 1] \cdot \mathbb{P}[N_t^T \geq 1] \mathbb{P}[\tau \in dt] \\ &= \int_0^\infty (1 - e^{-\lambda_P t}) \cdot (1 - e^{-\lambda_T t}) \cdot \lambda_A e^{-\lambda_A t} dt \\ &= \int_0^\infty \lambda_A e^{-\lambda_A t} dt - \int_0^\infty \lambda_A e^{-(\lambda_A + \lambda_P)t} dt \\ &\quad - \int_0^\infty \lambda_A e^{-(\lambda_A + \lambda_T)t} dt + \int_0^\infty \lambda_A e^{-(\lambda_A + \lambda_P + \lambda_T)t} dt \\ &= 1 - \frac{\lambda_A}{\lambda_A + \lambda_P} - \frac{\lambda_A}{\lambda_A + \lambda_T} + \frac{\lambda_A}{\lambda_A + \lambda_P + \lambda_T} \\ &= 1 - \frac{1}{201} - \frac{1}{101} + \frac{1}{301}. \end{aligned}$$

(With some effort, you can validate that the two expressions are equal.)